

PROJECT READINESS AUDIT

Detection and Analysis of Software Aging in Linux Kernel

CLI INSTRUCTION: Upload this document to the Gemini CLI. Ask: "Review my current codebase and logs against these technical requirements. Identify missing modules or experimental gaps."

1. Technical Implementation Checklist

Module	Required Functionality
Data Collector	Python daemon monitoring <i>/proc/meminfo</i> , <i>/proc/slabinfo</i> , and <i>/proc/buddyinfo</i> .
Detection Engine	HNSW-based or benchmark-driven logic to identify resource exhaustion patterns.
Rejuvenation	Automated triggers for cache clearing or process redistribution based on aging thresholds.

2. Experimental Validation Requirements

The project is considered "Ready" only if the following data points are present in the logs:

- **Accelerated Life Tests (ALT):** Evidence of system behavior under sustained artificial stress.
- **Fragmentation Metrics:** Quantitative analysis of memory fragmentation (Σext_frag) vs. time.
- **Predictive Accuracy:** Comparison between predicted Time-to-Failure (TTF) and actual system degradation.

3. CLI & Visualization Specs

Component	Requirement
Dashboard/CLI	Real-time display of "Kernel Health Score" and fragmentation levels.
Reporting	Capability to export CSV/JSON summaries of aging sessions.

4. Final Integrity Check

- ☐ Codebase handles 24h+ monitoring without significant overhead.
- ☐ Rejuvenation triggers do not cause system instability or data loss.
- ☐ Baseline vs. Managed performance comparison documented.